

Project Overview and Goals:

My final project was an analysis of the correlation between Google search trends and subreddit posts between November 23, 2022 and March 2, 2023. Examining how information flows from media to the search browsers and vice versa is a good way to research how information spreads through different online platforms. The main goals of the project were to discover whether Reddit discussions predict or mirror Google search trends, measure which of the four subreddits correlated most strongly with trending searches, and to deliver an interactive Looker Studio report with time series and drill down views.

Dataset Description:

1 Google Trends Dataset

The first dataset chosen was the BigQuery publicly available Google Trends dataset. The dataset has tables that include the daily top 25 search terms for the US, the daily top 25 fastest-rising terms in the US, and the top 25 of each category globally. Each term has its rank, popularity score, week, refresh date, and geographic location stored in the dataset. All of the data is structured (tabular) and is stored in a columnar format, making it more efficient to use when performing SQL queries. The dataset is updated daily with a new set of top 25 terms, and each new addition has a 30 day period of being in the dataset before rolling over into the historical data, which holds the last five years of search terms.

2 Reddit Dataset (Kaggle)

To meet the ingestion requirement of the project, the second dataset decided on was a Kaggle dataset with scraped data from Reddit's top 4 most popular subreddits: r/technology, r/worldnews, r/entertainment and r/sports. Each entry has the post title, author, url, post, and post date. Originally, the project included a feature using Cloud Composer to orchestrate the streaming data for the Google Trends dataset, but due to the non-streaming Kaggle dataset, implementing this feature did not happen.

Description of Pipeline Workflow and Services Used:

The first step of the project was ingesting the Reddit dataset into a Google Storage bucket that was created for the project. Then Dataproc was used to create a cluster to process the cleaning of the Reddit dataset. Next, BigQuery was used to extract keywords from the subreddit posts, create tables for visualizing in future steps, and to calculate the correlation scores between the

keywords and the trending terms from the Google dataset. Finally, the new tables were utilized to implement visualizations in Looker Studio.

Data Processing Steps:

1 Data Ingestion

The four Reddit CSV files were downloaded locally and then uploaded to the Google Cloud Storage bucket created for the project. The bucket is organized with the raw/ prefix for the original files and any transformed outputs were stored under the processed/ prefix.

2 PySpark Cleaning Job (Dataproc)

A Dataproc cluster was created with two worker nodes and a PySpark job for cleaning the data. The cleaning script dropped rows with null values, removed duplicates using the post id, normalized the column formats, and wrote the cleaned output in the Parquet format.

3 BigQuery Loading and Partitioning

The processed Parquet files were then loaded into BigQuery using the bq load command. They were then partitioned based on date and which subreddit the entry belonged to, so that queries are able to navigate and execute more efficiently.

4 Keyword Extraction

A SQL query was written to extract words from each Reddit post title using SPLIT and UNNEST functions, applying REGEXP_REPLACE to exclude non-alphanumeric characters. Words shorter than five characters and a custom list of common stop words were filtered out. The resulting keyword table (reddit_keywords) stored each keyword alongside its subreddit, post date, mention count, and daily rank within its subreddit.

5 Cross-Table Join with Google Trends

The reddit_keywords table was joined against the big-query-public-data.google_trends.top_terms table on two conditions: the keywords matching the Google trending term, and the Reddit post date falling within the same 7-day period as the Google Trends week. The result was stored in the reddit_vs_google_trends table, which forms the foundation of all downstream analysis.

Results:

1 Overall Results

Table/View	Description
reddit_posts_partitioned	Cleaned Reddit posts partitioned by post_date, clustered by subreddit
reddit_keywords	Extracted keywords per post with mention_count and daily rank
reddit_vs_google_trends	Matched Reddit keywords against Google Trends terms with 7-day windows
subreddit_correlation_scores	Aggregate correlation percentage per subreddit
weekly_trend_alignment	Weekly matching terms, Reddit mentions, and avg Google score by subreddit

2 Correlation Scores

The key result of the project was the subreddit correlation score, which was calculated as the percentage of a subreddit's extracted keywords that matched a Google trending term within the same week.

Subreddit	Matched	Correlation
r/sports	16,800	0.7186
r/worldnews	63,000	0.4020
r/entertainment	3,570	0.0868
r/technology	4,410	0.0503

r/sports demonstrated the strongest correlation at 0.7186, indicating that sports events drive simultaneous Reddit discussion and Google searches. r/worldnews followed at 0.4020, confirming that breaking news topics consistently cross platforms. r/entertainment and r/technology showed much lower correlation, suggesting these communities discuss niche or enthusiast topics that rarely reach mainstream search volumes.

Visualizations and Dashboard:

The final Looker Studio dashboard was connected directly to the BigQuery tables. The dashboard includes six interactive components.

1 Scorecard tiles

Four scorecards at the top of the dashboard display the correlation percentage for each subreddit at a glance.

2 Correlation bar chart

A vertical bar chart using the `subreddit_correlation_scores` table, showing each subreddit's overall correlation percentage sorted from highest to lowest.

3 Weekly time series

A line chart for the `weekly_correlation_summary` view showing total Reddit mentions and average Google score per week. This reveals whether Reddit activity and Google search interest rise and fall together over time.

4 Trending terms bar chart

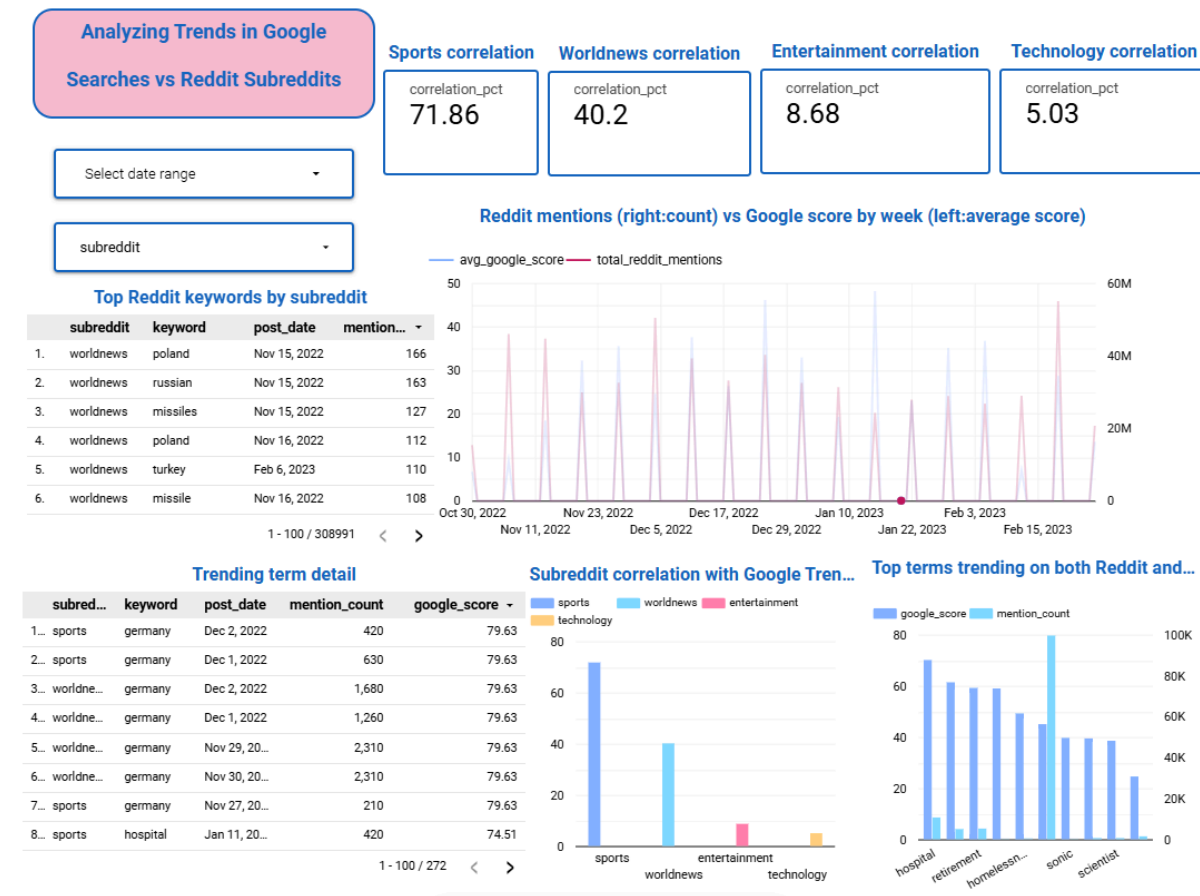
A bar chart from the `reddit_vs_google_trends` table showing the top keywords that appeared in both Reddit and Google Trends, displaying both Reddit mention count and Google score.

5 Drill-down detail tables

Sortable and filterable tables for examining top keywords for each subreddit and top trending search terms.

6 Range controls and subreddit filters

Dashboard includes a date range control and a subreddit dropdown filter, both configured to apply across all data sources in the report, enabling fully interactive cross-filtering.



Challenges Encountered and How They Were Resolved:

1 Cloud Shell Disk Space

When attempting to install the Kaggle CLI in Google Cloud Shell, the installation failed with a “No space left on device” error. Cloud Shell only provides 5GB of home directory storage, and the package dependencies consume the remaining space. The resolution was to download the dataset directly from the Kaggle website locally and upload the zip file to Google Cloud Storage using the browser-based GCS console, bypassing Cloud Shell entirely.

2 Dataproc Disk Quota

Creating the initial Dataproc cluster failed with an INVALID_ARGUMENT error due to insufficient DISKS_TOTAL_GB quota. The default Dataproc configuration requested 3,000GB of disk across the cluster, exceeding the project’s quota. The resolution was to explicitly set the worker and master book disk sizes to 100GB each using the --worker-boot-disk-size and --master-boot-disk-size flags, reducing total disk usage to 300GB.

Lessons Learned:

The most significant lessons learned were figuring out how to build a full Google Cloud pipeline and adjusting the configuration of the pipeline to the level of work needed for a smaller project. This also helps with cost management by using only the resources required for the work completed. When looking through the results, it is also clear that using single word tokens lacks contextual meaning. For example, a phrase like “World Cup” split into two words might not be fully captured by the analysis.

Potential Next Steps:

With additional time and resources, I would replace the static Kaggle Reddit dataset with an API that pulls live data from Reddit, which would eliminate the date range restrictions and allow real-time analysis of cross-platform trends. I would also move over to phrase-level keyword extraction. This would allow the analysis to capture multi-word trending terms. These pipeline improvements would allow for better, more expansive and relevant analysis.